

NKPC HSA Bayesian State-Space Estimation: Unemployment gap

June 22, 2026

1 Data and Transformations

Raw data are read from `data/raw/`. Processed model-ready data are written to `data/processed/`. The HSA competition aggregator convention is $N_t = 100 \log(N_t^{level})$ unless a run metadata file explicitly states that the series was already transformed.

2 Model Specifications

The CES benchmark is

$$\pi_t = \alpha\pi_{t-1} + (1 - \alpha)E_t\pi_{t+1} + \kappa x_t + e_t.$$

The dynamic HSA specification is

$$\pi_t = \alpha\pi_{t-1} + (1 - \alpha)E_t\pi_{t+1} + \kappa x_t - \theta\hat{N}_t + e_t.$$

The steady HSA specification is

$$\pi_t = \alpha\pi_{t-1} + (1 - \alpha)E_t\pi_{t+1} + (\kappa_0 + \delta\bar{N}_t)x_t + e_t.$$

The both-effects HSA specification is

$$\pi_t = \alpha\pi_{t-1} + (1 - \alpha)E_t\pi_{t+1} + (\kappa_0 + \delta\bar{N}_t)x_t - (\theta_0 + \gamma\bar{N}_t)\hat{N}_t + e_t.$$

The maintained HSA decomposition is

$$N_t = \hat{N}_t + \bar{N}_t, \quad \hat{N}_t = \rho_1\hat{N}_{t-1} + \rho_2\hat{N}_{t-2} + u_t, \quad \bar{N}_t = n + \bar{N}_{t-1} + \epsilon_t.$$

3 Unit and Covariance Conventions

Priors for κ , κ_0 , and δ are specified in physical units. Legacy Gibbs samplers internally use `KAPPA_SCALE=100` when the design matrix contains $x_t/100$. All posterior draws, tables, figures, SDDR calculations, and model-comparison summaries are reported in physical units. The default dynamic-HSA shock covariance allows only (e_t, ζ_t) correlation; diagonal and historical full covariance modes are available only through explicit run metadata.

4 Prior Specifications

Baseline, weak, and tight priors are stored in `configs/`. Robustness tables below summarize whether conclusions for δ , θ , and γ change across these choices.

5 1. Posterior Mean Coefficients

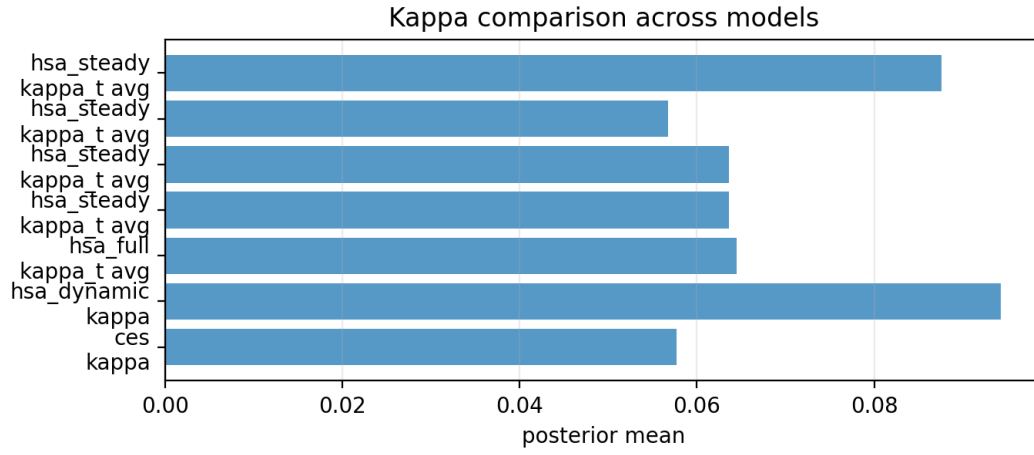
This section reports posterior means and credible intervals for the main scalar coefficients in each estimated run.

run	model	parameter	posterior_mean	posterior_sd	ci.2.5	ci.97.5
ces_unemployment_gap_baseline.20260622.131739	ces	alpha	0.6427	0.0616	0.5494	0.7623
ces_unemployment_gap_baseline.20260622.131739	ces	kappa	0.0577	0.0369	0.0063	0.1398
ces_unemployment_gap_baseline.20260622.131739	ces	phi.1	0.9533	0.0306	0.8906	0.9957
ces_unemployment_gap_baseline.20260622.131739	ces	lambda_ez	0.0276	0.0971	-0.1368	0.1816
hsa_dynamic_unemployment_gap_baseline.20260622.131740	hsa_dynamic	alpha	0.5917	0.0454	0.5213	0.6629
hsa_dynamic_unemployment_gap_baseline.20260622.131740	hsa_dynamic	kappa	0.0943	0.0348	0.0443	0.1599
hsa_dynamic_unemployment_gap_baseline.20260622.131740	hsa_dynamic	theta	0.0117	0.0033	0.0063	0.0185
hsa_dynamic_unemployment_gap_baseline.20260622.131740	hsa_dynamic	phi.1	0.9545	0.0264	0.9117	0.9997
hsa_dynamic_unemployment_gap_baseline.20260622.131740	hsa_dynamic	rho.1	1.8723	0.0418	1.7904	1.9412
hsa_dynamic_unemployment_gap_baseline.20260622.131740	hsa_dynamic	rho.2	-0.8759	0.0417	-0.9454	-0.7923
hsa_dynamic_unemployment_gap_baseline.20260622.131740	hsa_dynamic	n	0.0007	0.0562	-0.0913	0.0972
hsa_full_unemployment_gap_baseline.20260622.131742	hsa_full	alpha	0.6335	0.0557	0.5330	0.7437
hsa_full_unemployment_gap_baseline.20260622.131742	hsa_full	kappa.0	-0.0591	0.1845	-0.3698	0.2173
hsa_full_unemployment_gap_baseline.20260622.131742	hsa_full	delta	0.0006	0.0010	-0.0008	0.0026
hsa_full_unemployment_gap_baseline.20260622.131742	hsa_full	theta.0	0.0261	0.2315	-0.3917	0.4344
hsa_full_unemployment_gap_baseline.20260622.131742	hsa_full	gamma	0.0000	0.0011	-0.0019	0.0020
hsa_full_unemployment_gap_baseline.20260622.131742	hsa_full	phi.1	0.9503	0.0249	0.9149	1.0017
hsa_full_unemployment_gap_baseline.20260622.131742	hsa_full	rho.1	1.1937	0.1128	1.0049	1.3924
hsa_full_unemployment_gap_baseline.20260622.131742	hsa_full	rho.2	-0.2715	0.1195	-0.4719	-0.0685
hsa_full_unemployment_gap_baseline.20260622.131742	hsa_full	n	-0.0903	0.0589	-0.1810	0.0366
hsa_full_unemployment_gap_baseline.20260622.131742	hsa_full	lambda_ez	0.0194	0.1184	-0.1696	0.2203
hsa_steady_unemployment_gap_baseline.20260622.131739	hsa_steady	alpha	0.6577	0.0553	0.5494	0.7578
hsa_steady_unemployment_gap_baseline.20260622.131739	hsa_steady	kappa.0	0.0757	0.2048	-0.3673	0.3665
hsa_steady_unemployment_gap_baseline.20260622.131739	hsa_steady	delta	-0.0001	0.0010	-0.0017	0.0021
hsa_steady_unemployment_gap_baseline.20260622.131739	hsa_steady	phi.1	0.9533	0.0254	0.9065	0.9897
hsa_steady_unemployment_gap_baseline.20260622.131739	hsa_steady	rho.1	1.8134	0.0481	1.7138	1.9009
hsa_steady_unemployment_gap_baseline.20260622.131739	hsa_steady	rho.2	-0.8168	0.0481	-0.9047	-0.7165
hsa_steady_unemployment_gap_baseline.20260622.131739	hsa_steady	n	-0.0609	0.0738	-0.1749	0.0646
hsa_steady_unemployment_gap_baseline.20260622.131739	hsa_steady	lambda_ez	0.0233	0.0967	-0.1479	0.1884
hsa_steady_unemployment_gap_baseline.20260622.132052	hsa_steady	alpha	0.6577	0.0553	0.5494	0.7578
hsa_steady_unemployment_gap_baseline.20260622.132052	hsa_steady	kappa.0	0.0757	0.2048	-0.3673	0.3665
hsa_steady_unemployment_gap_baseline.20260622.132052	hsa_steady	delta	-0.0001	0.0010	-0.0017	0.0021
hsa_steady_unemployment_gap_baseline.20260622.132052	hsa_steady	phi.1	0.9533	0.0254	0.9065	0.9897
hsa_steady_unemployment_gap_baseline.20260622.132052	hsa_steady	rho.1	1.8134	0.0481	1.7138	1.9009
hsa_steady_unemployment_gap_baseline.20260622.132052	hsa_steady	rho.2	-0.8168	0.0481	-0.9047	-0.7165
hsa_steady_unemployment_gap_baseline.20260622.132052	hsa_steady	n	-0.0609	0.0738	-0.1749	0.0646
hsa_steady_unemployment_gap_baseline.20260622.132052	hsa_steady	lambda_ez	0.0233	0.0967	-0.1479	0.1884
hsa_steady_unemployment_gap_tight.20260622.132056	hsa_steady	alpha	0.6170	0.0560	0.5139	0.7149
hsa_steady_unemployment_gap_tight.20260622.132056	hsa_steady	kappa.0	0.0459	0.1058	-0.1360	0.2705
hsa_steady_unemployment_gap_tight.20260622.132056	hsa_steady	delta	0.0001	0.0005	-0.0009	0.0009
hsa_steady_unemployment_gap_tight.20260622.132056	hsa_steady	phi.1	0.9366	0.0242	0.8903	0.9726
hsa_steady_unemployment_gap_tight.20260622.132056	hsa_steady	rho.1	0.5943	0.1034	0.3941	0.7615
hsa_steady_unemployment_gap_tight.20260622.132056	hsa_steady	rho.2	-0.3245	0.1371	-0.5107	-0.0126
hsa_steady_unemployment_gap_tight.20260622.132056	hsa_steady	n	-0.0195	0.0443	-0.0959	0.0561
hsa_steady_unemployment_gap_tight.20260622.132056	hsa_steady	lambda_ez	0.0154	0.1119	-0.1602	0.1919
hsa_steady_unemployment_gap_weak.20260622.132053	hsa_steady	alpha	0.6432	0.0474	0.5567	0.7306
hsa_steady_unemployment_gap_weak.20260622.132053	hsa_steady	kappa.0	-0.6887	0.3050	-1.1713	-0.1801
hsa_steady_unemployment_gap_weak.20260622.132053	hsa_steady	delta	0.0041	0.0016	0.0015	0.0066
hsa_steady_unemployment_gap_weak.20260622.132053	hsa_steady	phi.1	0.9550	0.0223	0.9171	0.9996
hsa_steady_unemployment_gap_weak.20260622.132053	hsa_steady	rho.1	1.9052	0.0385	1.8457	1.9727
hsa_steady_unemployment_gap_weak.20260622.132053	hsa_steady	rho.2	-0.9085	0.0383	-0.9745	-0.8494
hsa_steady_unemployment_gap_weak.20260622.132053	hsa_steady	n	-0.2724	0.0842	-0.4217	-0.1244
hsa_steady_unemployment_gap_weak.20260622.132053	hsa_steady	lambda_ez	-0.0001	0.1042	-0.2115	0.1681

6 2. Kappa Comparison Across Models

For CES models this is κ . For steady or full HSA models the table also reports κ_0 , δ , and the average path of $\kappa_t = \kappa_0 + \delta \bar{N}_t$ when available.

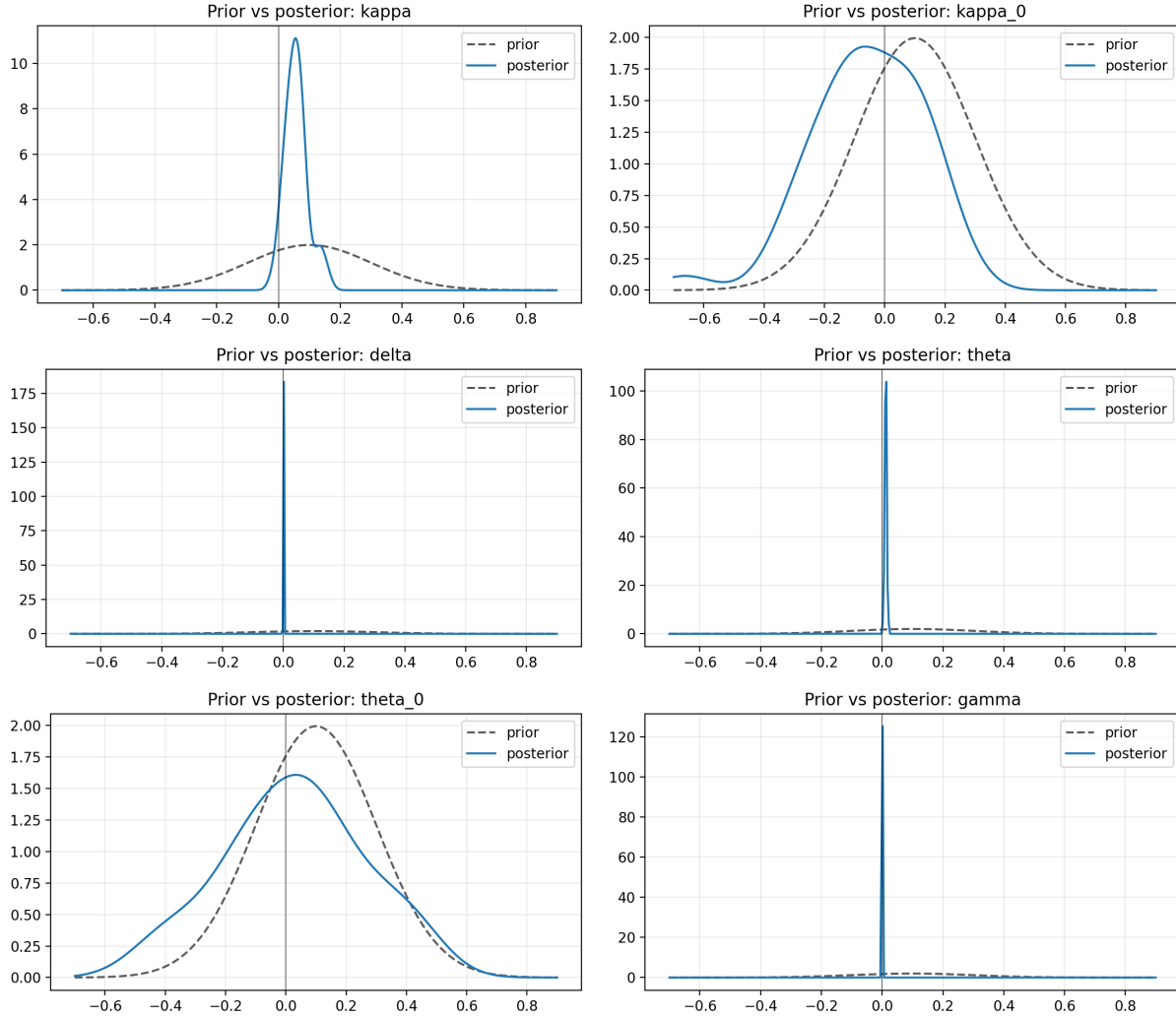
run	model	kappa_mean	kappa_ci.2.5	kappa_ci.97.5	kappa_0_mean	kappa_0_ci.2.5	kappa_0_ci.97.5	delta_mean	delta_ci.2.5	delta_ci.97.5	kappa_t_overall_mean	kappa_t_start_mean	kappa_t_end_mean
ces_unemployment_gap_baseline.20260622.131739	ces	0.0577	0.0063	0.1398	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
hsa_dynamic_unemployment_gap_baseline.20260622.131740	hsa_dynamic	0.0943	0.0443	0.1599	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
hsa_full_unemployment_gap_baseline.20260622.131742	hsa_full	NaN	NaN	NaN	-0.0591	-0.3698	0.2173	0.0006	-0.0008	0.0026	0.0044	0.0020	0.0485
hsa_steady_unemployment_gap_baseline.20260622.131739	hsa_steady	NaN	NaN	NaN	0.0757	-0.3673	0.3665	-0.0001	-0.0017	0.0021	0.0636	0.0638	0.0629
hsa_steady_unemployment_gap_tight.20260622.132056	hsa_steady	NaN	NaN	NaN	0.0757	-0.3673	0.3665	-0.0001	-0.0017	0.0021	0.0636	0.0638	0.0629
hsa_steady_unemployment_gap_weak.20260622.132053	hsa_steady	NaN	NaN	NaN	0.0459	-1.1713	-0.1801	0.0041	0.0015	0.0066	0.0567	0.0564	0.0552



7 3. Time-Varying Coefficient Paths

The main time-varying coefficients are κ_t and, in the both-effects model, $\theta_t = \theta_0 + \gamma \bar{N}_t$.

run	model	restriction	prior_mean	prior_sd	sddr_bf01
ces_unemployment_gap_baseline_20260622_131739	ces	kappa=0	0.1000	0.2000	2.0389
hsa_dynamic_unemployment_gap_baseline_20260622_131740	hsa_dynamic	kappa=0	0.1000	0.2000	0.1520
hsa_dynamic_unemployment_gap_baseline_20260622_131740	hsa_dynamic	theta=0	0.1000	0.2000	0.5418
hsa_full_unemployment_gap_baseline_20260622_131742	hsa_full	kappa_0=0	0.1000	0.2000	1.0683
hsa_full_unemployment_gap_baseline_20260622_131742	hsa_full	delta=0	0.1000	0.2000	187.3843
hsa_full_unemployment_gap_baseline_20260622_131742	hsa_full	theta_0=0	0.1000	0.2000	0.9029
hsa_full_unemployment_gap_baseline_20260622_131742	hsa_full	gamma=0	0.1000	0.2000	204.9817
hsa_steady_unemployment_gap_baseline_20260622_131739	hsa_steady	kappa_0=0	0.1000	0.2000	0.9013
hsa_steady_unemployment_gap_baseline_20260622_131739	hsa_steady	delta=0	0.1000	0.2000	201.9449
hsa_steady_unemployment_gap_baseline_20260622_132052	hsa_steady	kappa_0=0	0.1000	0.2000	0.9013
hsa_steady_unemployment_gap_baseline_20260622_132052	hsa_steady	delta=0	0.1000	0.2000	201.9449
hsa_steady_unemployment_gap_tight_20260622_132056	hsa_steady	kappa_0=0	0.1000	0.2000	1.9888
hsa_steady_unemployment_gap_tight_20260622_132056	hsa_steady	delta=0	0.1000	0.2000	421.1822
hsa_steady_unemployment_gap_weak_20260622_132053	hsa_steady	kappa_0=0	0.1000	0.2000	0.0928
hsa_steady_unemployment_gap_weak_20260622_132053	hsa_steady	delta=0	0.1000	0.2000	8.4500



10 Diagnostics and Robustness

run	model	data_spec	prior_spec	parameter	mean	sd	hdi_2.5%	hdi_97.5%	mcmc_mean	ess_bulk	ess_tail	r_hat	warning
ces.unemployment_gap.baseline.20260622.131739	ces	unemployment_gap	baseline	alpha	0.6427	0.0616	0.5494	0.7623	NaN	NaN	NaN	NaN	NaN
ces.unemployment_gap.baseline.20260622.131739	ces	unemployment_gap	baseline	kappa	0.0577	0.0369	0.0063	0.1398	NaN	NaN	NaN	NaN	NaN
ces.unemployment_gap.baseline.20260622.131739	ces	unemployment_gap	baseline	phi.1	0.9533	0.0306	0.8906	0.9957	NaN	NaN	NaN	NaN	NaN
ces.unemployment_gap.baseline.20260622.131739	ces	unemployment_gap	baseline	lambda_ez	0.0276	0.0071	-0.1368	0.1816	NaN	NaN	NaN	NaN	NaN
ces.unemployment_gap.baseline.20260622.131739	ces	unemployment_gap	baseline	rho	0.0223	0.0880	-0.1154	0.1679	NaN	NaN	NaN	NaN	NaN
ces.unemployment_gap.baseline.20260622.131739	ces	unemployment_gap	baseline	sigma_e	0.6533	0.0435	0.5796	0.7038	NaN	NaN	NaN	NaN	NaN
ces.unemployment_gap.baseline.20260622.131739	ces	unemployment_gap	baseline	sigma_zeta	0.6014	0.0364	0.5422	0.6628	NaN	NaN	NaN	NaN	NaN
hsa.dynamic.unemployment_gap.baseline.20260622.131740	hsa.dynamic	unemployment_gap	baseline	alpha	0.5917	0.0454	0.5213	0.6629	NaN	NaN	NaN	NaN	NaN
hsa.dynamic.unemployment_gap.baseline.20260622.131740	hsa.dynamic	unemployment_gap	baseline	kappa	0.0943	0.0348	0.0443	0.1599	NaN	NaN	NaN	NaN	NaN
hsa.dynamic.unemployment_gap.baseline.20260622.131740	hsa.dynamic	unemployment_gap	baseline	theta	0.0117	0.0033	0.0063	0.0185	NaN	NaN	NaN	NaN	NaN
hsa.dynamic.unemployment_gap.baseline.20260622.131740	hsa.dynamic	unemployment_gap	baseline	phi.1	0.9545	0.0264	0.9117	0.9997	NaN	NaN	NaN	NaN	NaN
hsa.dynamic.unemployment_gap.baseline.20260622.131740	hsa.dynamic	unemployment_gap	baseline	rho.1	1.8723	0.0418	1.7904	1.9412	NaN	NaN	NaN	NaN	NaN
hsa.dynamic.unemployment_gap.baseline.20260622.131740	hsa.dynamic	unemployment_gap	baseline	rho.2	-0.8759	0.0417	-0.9454	-0.7923	NaN	NaN	NaN	NaN	NaN
hsa.dynamic.unemployment_gap.baseline.20260622.131740	hsa.dynamic	unemployment_gap	baseline	n	0.0007	0.0562	-0.0913	0.0972	NaN	NaN	NaN	NaN	NaN
hsa.dynamic.unemployment_gap.baseline.20260622.131740	hsa.dynamic	unemployment_gap	baseline	sigma_e	0.6197	0.0368	0.5713	0.6977	NaN	NaN	NaN	NaN	NaN
hsa.dynamic.unemployment_gap.baseline.20260622.131740	hsa.dynamic	unemployment_gap	baseline	sigma_zeta	0.5807	0.0348	0.5309	0.6649	NaN	NaN	NaN	NaN	NaN
hsa.dynamic.unemployment_gap.baseline.20260622.131740	hsa.dynamic	unemployment_gap	baseline	sigma_u	0.4183	0.0448	0.3466	0.4953	NaN	NaN	NaN	NaN	NaN
hsa.dynamic.unemployment_gap.baseline.20260622.131740	hsa.dynamic	unemployment_gap	baseline	sigma_eps	0.2555	0.0262	0.2131	0.3121	NaN	NaN	NaN	NaN	NaN
hsa.dynamic.unemployment_gap.baseline.20260622.131740	hsa.dynamic	unemployment_gap	baseline	sigma_N	0.2888	0.0239	0.2378	0.3269	NaN	NaN	NaN	NaN	NaN
hsa.full.unemployment_gap.baseline.20260622.131742	hsa.full	unemployment_gap	baseline	alpha	0.6335	0.0557	0.5330	0.7437	NaN	NaN	NaN	NaN	NaN
hsa.full.unemployment_gap.baseline.20260622.131742	hsa.full	unemployment_gap	baseline	kappa.0	-0.0591	0.1845	-0.3698	0.2173	NaN	NaN	NaN	NaN	NaN
hsa.full.unemployment_gap.baseline.20260622.131742	hsa.full	unemployment_gap	baseline	delta	0.0006	0.0010	-0.0008	0.0026	NaN	NaN	NaN	NaN	NaN
hsa.full.unemployment_gap.baseline.20260622.131742	hsa.full	unemployment_gap	baseline	theta.0	0.0261	0.2315	-0.3917	0.4344	NaN	NaN	NaN	NaN	NaN
hsa.full.unemployment_gap.baseline.20260622.131742	hsa.full	unemployment_gap	baseline	gamma	0.0000	0.0011	-0.0019	0.0020	NaN	NaN	NaN	NaN	NaN
hsa.full.unemployment_gap.baseline.20260622.131742	hsa.full	unemployment_gap	baseline	kappa.t.mean	0.0644	0.0446	-0.0093	0.1610	NaN	NaN	NaN	NaN	NaN
hsa.full.unemployment_gap.baseline.20260622.131742	hsa.full	unemployment_gap	baseline	kappa.t.start	0.0620	0.0426	-0.0060	0.1559	NaN	NaN	NaN	NaN	NaN
hsa.full.unemployment_gap.baseline.20260622.131742	hsa.full	unemployment_gap	baseline	kappa.t.mid	0.0750	0.0571	-0.0120	0.1899	NaN	NaN	NaN	NaN	NaN
hsa.full.unemployment_gap.baseline.20260622.131742	hsa.full	unemployment_gap	baseline	kappa.t.end	0.0485	0.0361	-0.0156	0.1206	NaN	NaN	NaN	NaN	NaN
hsa.full.unemployment_gap.baseline.20260622.131742	hsa.full	unemployment_gap	baseline	theta.t.mean	0.0334	0.0284	-0.0261	0.0930	NaN	NaN	NaN	NaN	NaN
hsa.full.unemployment_gap.baseline.20260622.131742	hsa.full	unemployment_gap	baseline	theta.t.start	0.0333	0.0303	-0.0320	0.0925	NaN	NaN	NaN	NaN	NaN
hsa.full.unemployment_gap.baseline.20260622.131742	hsa.full	unemployment_gap	baseline	theta.t.mid	0.0342	0.0268	-0.0055	0.0962	NaN	NaN	NaN	NaN	NaN
hsa.full.unemployment_gap.baseline.20260622.131742	hsa.full	unemployment_gap	baseline	theta.t.end	0.0319	0.0475	-0.0717	0.1030	NaN	NaN	NaN	NaN	NaN
hsa.full.unemployment_gap.baseline.20260622.131742	hsa.full	unemployment_gap	baseline	phi.1	0.9503	0.0249	0.9149	1.0017	NaN	NaN	NaN	NaN	NaN
hsa.full.unemployment_gap.baseline.20260622.131742	hsa.full	unemployment_gap	baseline	rho.1	1.1937	0.1128	1.0049	1.3924	NaN	NaN	NaN	NaN	NaN
hsa.full.unemployment_gap.baseline.20260622.131742	hsa.full	unemployment_gap	baseline	rho.2	-0.2715	0.1195	-0.4719	-0.0685	NaN	NaN	NaN	NaN	NaN
hsa.full.unemployment_gap.baseline.20260622.131742	hsa.full	unemployment_gap	baseline	n	-0.0903	0.0589	-0.1810	0.0366	NaN	NaN	NaN	NaN	NaN
hsa.full.unemployment_gap.baseline.20260622.131742	hsa.full	unemployment_gap	baseline	lambda_ez	0.0194	0.1184	-0.1696	0.2203	NaN	NaN	NaN	NaN	NaN
hsa.full.unemployment_gap.baseline.20260622.131742	hsa.full	unemployment_gap	baseline	rho	0.0176	0.1058	-0.1670	0.2092	NaN	NaN	NaN	NaN	NaN
hsa.full.unemployment_gap.baseline.20260622.131742	hsa.full	unemployment_gap	baseline	sigma_e	0.6701	0.0475	0.5896	0.7371	NaN	NaN	NaN	NaN	NaN
hsa.full.unemployment_gap.baseline.20260622.131742	hsa.full	unemployment_gap	baseline	sigma_zeta	0.5921	0.0390	0.5300	0.6779	NaN	NaN	NaN	NaN	NaN
hsa.full.unemployment_gap.baseline.20260622.131742	hsa.full	unemployment_gap	baseline	sigma_u	0.6961	0.0508	0.5995	0.8435	NaN	NaN	NaN	NaN	NaN
hsa.full.unemployment_gap.baseline.20260622.131742	hsa.full	unemployment_gap	baseline	sigma_eps	1.0594	0.0799	0.9265	1.1800	NaN	NaN	NaN	NaN	NaN
hsa.steady.unemployment_gap.baseline.20260622.131739	hsa.steady	unemployment_gap	baseline	alpha	0.6577	0.0553	0.5494	0.7578	NaN	NaN	NaN	NaN	NaN
hsa.steady.unemployment_gap.baseline.20260622.131739	hsa.steady	unemployment_gap	baseline	kappa.0	0.0757	0.2048	-0.3673	0.3665	NaN	NaN	NaN	NaN	NaN
hsa.steady.unemployment_gap.baseline.20260622.131739	hsa.steady	unemployment_gap	baseline	delta	-0.0001	0.0010	-0.0017	0.0021	NaN	NaN	NaN	NaN	NaN
hsa.steady.unemployment_gap.baseline.20260622.131739	hsa.steady	unemployment_gap	baseline	kappa.t.mean	0.0636	0.0331	0.0046	0.1185	NaN	NaN	NaN	NaN	NaN
hsa.steady.unemployment_gap.baseline.20260622.131739	hsa.steady	unemployment_gap	baseline	kappa.t.start	0.0638	0.0352	0.0002	0.1186	NaN	NaN	NaN	NaN	NaN
hsa.steady.unemployment_gap.baseline.20260622.131739	hsa.steady	unemployment_gap	baseline	kappa.t.mid	0.0636	0.0328	0.0041	0.1185	NaN	NaN	NaN	NaN	NaN
hsa.steady.unemployment_gap.baseline.20260622.131739	hsa.steady	unemployment_gap	baseline	kappa.t.end	0.0629	0.0325	-0.0007	0.1183	NaN	NaN	NaN	NaN	NaN
hsa.steady.unemployment_gap.baseline.20260622.131739	hsa.steady	unemployment_gap	baseline	phi.1	0.9533	0.0254	0.9065	0.9897	NaN	NaN	NaN	NaN	NaN
hsa.steady.unemployment_gap.baseline.20260622.131739	hsa.steady	unemployment_gap	baseline	rho.1	1.8134	0.0481	1.7138	1.9009	NaN	NaN	NaN	NaN	NaN
hsa.steady.unemployment_gap.baseline.20260622.131739	hsa.steady	unemployment_gap	baseline	rho.2	-0.8168	0.0481	-0.9047	-0.7165	NaN	NaN	NaN	NaN	NaN
hsa.steady.unemployment_gap.baseline.20260622.131739	hsa.steady	unemployment_gap	baseline	n	-0.0609	0.0738	-0.1749	0.0646	NaN	NaN	NaN	NaN	NaN
hsa.steady.unemployment_gap.baseline.20260622.131739	hsa.steady	unemployment_gap	baseline	lambda_ez	0.0233	0.0967	-0.1479	0.1884	NaN	NaN	NaN	NaN	NaN
hsa.steady.unemployment_gap.baseline.20260622.131739	hsa.steady	unemployment_gap	baseline	rho	0.0197	0.0847	-0.1297	0.1700	NaN	NaN	NaN	NaN	NaN
hsa.steady.unemployment_gap.baseline.20260622.131739	hsa.steady	unemployment_gap	baseline	sigma_eta	0.6643	0.0415	0.6000	0.7319	NaN	NaN	NaN	NaN	NaN
hsa.steady.unemployment_gap.baseline.20260622.131739	hsa.steady	unemployment_gap	baseline	sigma_e	0.6668	0.0416	0.6003	0.7330	NaN	NaN	NaN	NaN	NaN
hsa.steady.unemployment_gap.baseline.20260622.131739	hsa.steady	unemployment_gap	baseline	sigma_zeta	0.5877	0.0383	0.5293	0.6718	NaN	NaN	NaN	NaN	NaN
hsa.steady.unemployment_gap.baseline.20260622.131739	hsa.steady	unemployment_gap	baseline	sigma_u	0.5017	0.0557	0.4005	0.5937	NaN	NaN	NaN	NaN	NaN
hsa.steady.unemployment_gap.baseline.20260622.131739	hsa.steady	unemployment_gap	baseline	sigma_eps	0.3803	0.0393	0.3130	0.4622	NaN	NaN	NaN	NaN	NaN
hsa.steady.unemployment_gap.baseline.20260622.131739	hsa.steady	unemployment_gap	baseline	sigma_N	0.3044	0.0312	0.2567	0.3591	NaN	NaN	NaN	NaN	NaN

data_spec	prior	parameter	mean	median	ci_2.5	ci_97.5	p_gt_0	p_lt_0	conclusion
unemployment_gap	baseline	alpha	0.6577	0.6608	0.5494	0.7578	1.0000	0.0000	NaN
unemployment_gap	baseline	kappa_0	0.0757	0.0893	-0.3673	0.3665	0.7000	0.3000	NaN
unemployment_gap	baseline	delta	-0.0001	-0.0001	-0.0017	0.0021	0.4750	0.5250	NaN
unemployment_gap	baseline	rho_1	1.8134	1.8142	1.7138	1.9009	1.0000	0.0000	NaN
unemployment_gap	baseline	rho_2	-0.8168	-0.8150	-0.9047	-0.7165	0.0000	1.0000	NaN
unemployment_gap	baseline	sigma_e	0.6668	0.6589	0.6003	0.7330	1.0000	0.0000	NaN
unemployment_gap	baseline	sigma_zeta	0.5877	0.5907	0.5293	0.6718	1.0000	0.0000	NaN
unemployment_gap	baseline	sigma_u	0.5017	0.5123	0.4005	0.5937	1.0000	0.0000	NaN
unemployment_gap	baseline	sigma_eps	0.3803	0.3806	0.3130	0.4622	1.0000	0.0000	NaN
unemployment_gap	baseline	sigma_N	0.3044	0.3028	0.2567	0.3591	1.0000	0.0000	NaN
unemployment_gap	weak	alpha	0.6432	0.6343	0.5567	0.7306	1.0000	0.0000	NaN
unemployment_gap	weak	kappa_0	-0.6887	-0.7053	-1.1713	-0.1801	0.0000	1.0000	NaN
unemployment_gap	weak	delta	0.0041	0.0045	0.0015	0.0066	1.0000	0.0000	NaN
unemployment_gap	weak	rho_1	1.9052	1.9079	1.8457	1.9727	1.0000	0.0000	NaN
unemployment_gap	weak	rho_2	-0.9085	-0.9110	-0.9745	-0.8494	0.0000	1.0000	NaN
unemployment_gap	weak	sigma_e	0.6305	0.6286	0.5643	0.6991	1.0000	0.0000	NaN
unemployment_gap	weak	sigma_zeta	0.5975	0.5975	0.5353	0.6788	1.0000	0.0000	NaN
unemployment_gap	weak	sigma_u	0.4045	0.4083	0.3273	0.4562	1.0000	0.0000	NaN
unemployment_gap	weak	sigma_eps	0.3022	0.2984	0.2446	0.3654	1.0000	0.0000	NaN
unemployment_gap	weak	sigma_N	0.2308	0.2245	0.1893	0.2702	1.0000	0.0000	NaN
unemployment_gap	tight	alpha	0.6170	0.6287	0.5139	0.7149	1.0000	0.0000	NaN
unemployment_gap	tight	kappa_0	0.0459	0.0441	-0.1360	0.2705	0.6500	0.3500	NaN
unemployment_gap	tight	delta	0.0001	0.0001	-0.0009	0.0009	0.6000	0.4000	NaN
unemployment_gap	tight	rho_1	0.5943	0.5906	0.3941	0.7615	1.0000	0.0000	NaN
unemployment_gap	tight	rho_2	-0.3245	-0.3362	-0.5107	-0.0126	0.0250	0.9750	NaN
unemployment_gap	tight	sigma_e	0.6662	0.6636	0.6126	0.7458	1.0000	0.0000	NaN
unemployment_gap	tight	sigma_zeta	0.5843	0.5805	0.5317	0.6505	1.0000	0.0000	NaN
unemployment_gap	tight	sigma_u	0.5610	0.5584	0.4782	0.6539	1.0000	0.0000	NaN
unemployment_gap	tight	sigma_eps	1.2545	1.2520	1.0921	1.4157	1.0000	0.0000	NaN
unemployment_gap	tight	sigma_N	0.5287	0.5229	0.4341	0.6206	1.0000	0.0000	NaN
unemployment_gap	all	delta_robustness	NaN	NaN	NaN	NaN	NaN	NaN	sensitive

note

Run scripts/05_period_robustness.py first.

11 Posterior Predictive Checks and Figures

Posterior predictive checks are generated after model runs.

12 Notes

WAIC and LOO are not used as primary criteria because the models contain latent states. Chib marginal likelihoods are reported only when the legacy ordinate calculation can be matched to the sampler units and supplied data.